

Questions 10.4 · 10.5 · 10.6 · 11.5

Logistic Regression and MANOVA Analysis

Q10.4

Admissions Data

Q10.5

Cholesterol Data

Q10.6

Depression Data

Q11.5

MANOVA

LR with GPA category (dummy-coded) + GMAT | Hand calculations + LR with dummy cholesterol levels | LR vs Discriminant Analysis (CASES as outcome) | Depression (C1-C20) + Phone (A1-A6) with LDA comparison

Tools: Python (statsmodels · sklearn · scipy · matplotlib · seaborn) | SAS (PROC LOGISTIC · PROC DISCRIM · PROC GLM)

Admissions Data — Logistic Regression Setup

Dataset: **ADMIS.DAT**

85 applicants: 31 admitted (status=1), 28 not-admitted (status=2).
Borderline (status=3) excluded — analysis uses only clear decisions.
New variable: admit = 0 (admitted), 1 (not-admitted).

GPA Category Recoding

GPA < 2.50 → Category 1 | 2.51–3.00 → Category 2
3.01–3.50 → Category 3 | > 3.50 → Category 4
Dummy coded with Category 1 as reference (3 dummies: GPA2, GPA3, GPA4).

Perfect Separation Note

GPA category perfectly separates the two classes:
All Cat 3 & 4 applicants were admitted; all Cat 1 were not admitted.
Maximum Likelihood estimates diverge — L2-penalized LR (C=1) used for stable coefficients.

GPA Category Split

Category	GPA Range	Admitted	Not Adm
1	< 2.50	0	15
2	2.51–3.00	1	13
3	3.01–3.50	22	0
4	> 3.50	8	0

Admissions — Model Results & Interpretation

98.3%

Model 1 Accuracy
(GPA only)

0.975

Model 1 AUC

96.6%

Model 2 Accuracy
(GPA+GMAT)

1.000

Model 2 AUC

Effect of GPA Category

GPA category is the dominant predictor — it alone achieves 98.3% accuracy. Coefficients for GPA3 and GPA4 are strongly negative (lower P(not-admitted)), confirming higher GPA strongly predicts admission. GPA2 slightly increases risk vs Category 1 baseline.

Effect of Adding GMAT

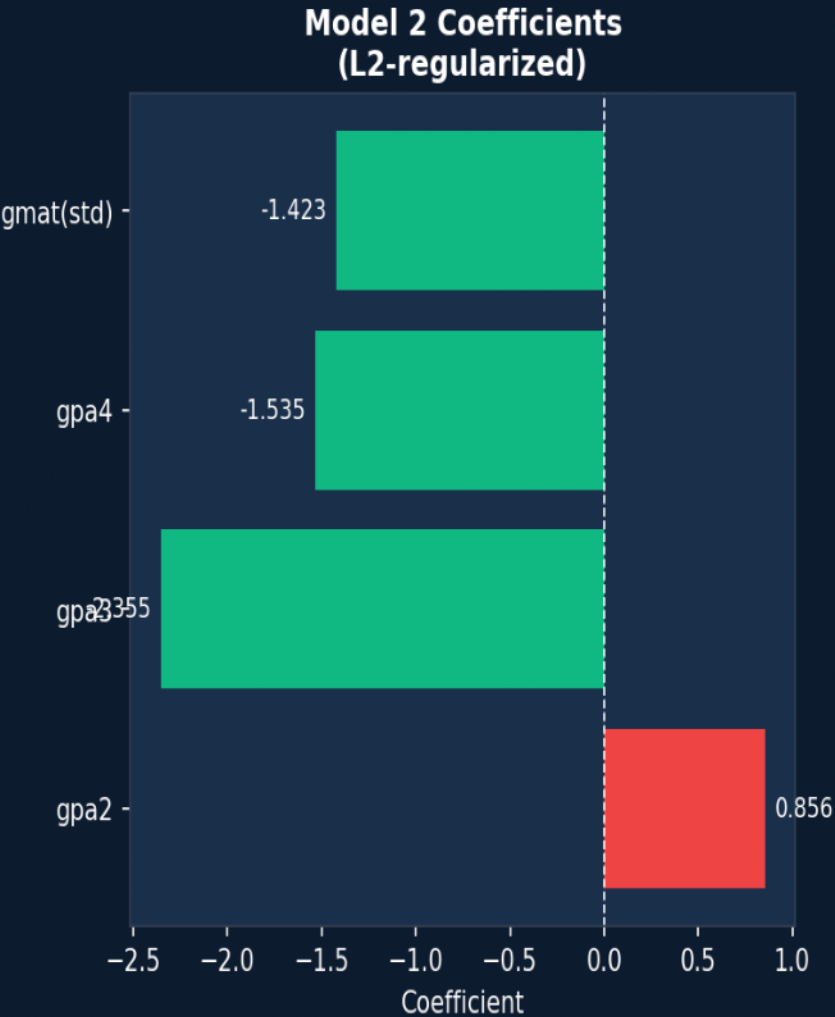
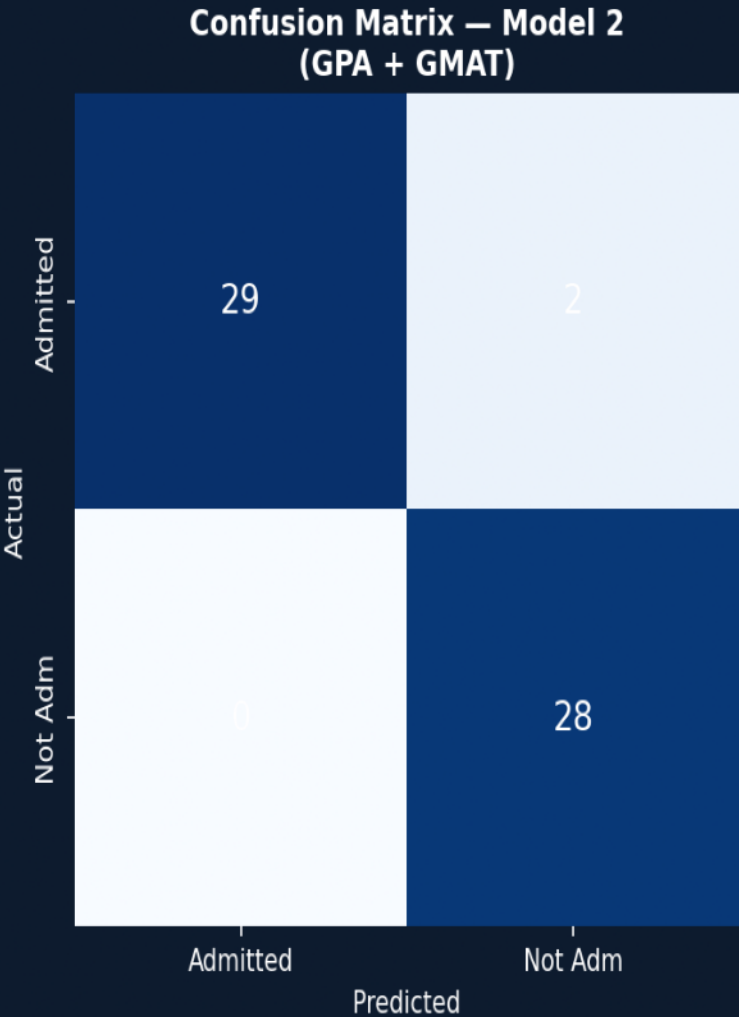
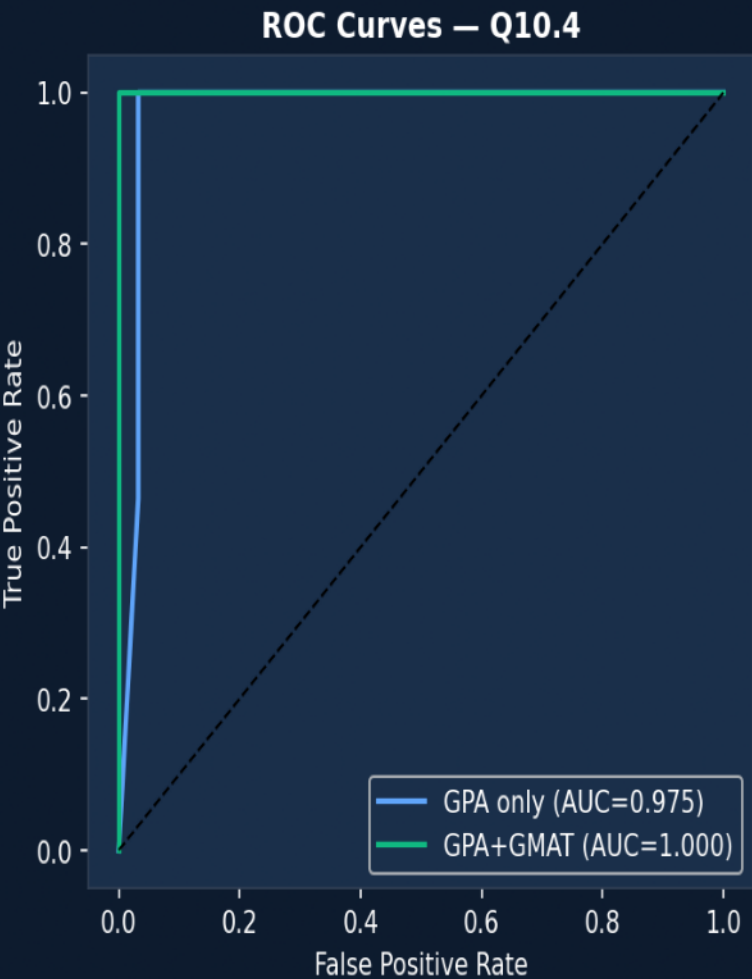
Model 2 adds GMAT score. AUC improves to 1.000 (perfect discrimination). GMAT coefficient is negative: higher GMAT → lower P(not-admitted). However, accuracy drops slightly (96.6%) due to 2 misclassifications. Overall, GMAT provides marginal improvement over GPA category alone.

Interpretation

GPA category alone provides near-perfect classification (98.3%), supporting it as the primary admissions criterion. Adding GMAT achieves a perfect AUC = 1.000, meaning the combined model perfectly ranks applicants by default risk. The GPA-only model is simpler and nearly as accurate. Both models confirm: higher GPA category → dramatically lower probability of rejection.

Admissions — Visualizations

ROC Curves, Confusion Matrix, and Coefficients



Cholesterol & Heart Disease — Hand Calculations

(a) Probabilities and Log-Odds:

114 males (40–65 yrs) classified by blood cholesterol and whether they developed heart disease.

	< 200	200-219	220-259	> 259
Present (HD)	6	10	30	45
Absent	5	6	5	7
Total	11	16	35	52
P(Disease)	0.545	0.625	0.857	0.865
Odds	1.200	1.667	6.000	6.429
Log-Odds	0.182	0.511	1.792	1.861

(b) Hand Logistic Regression Equation:

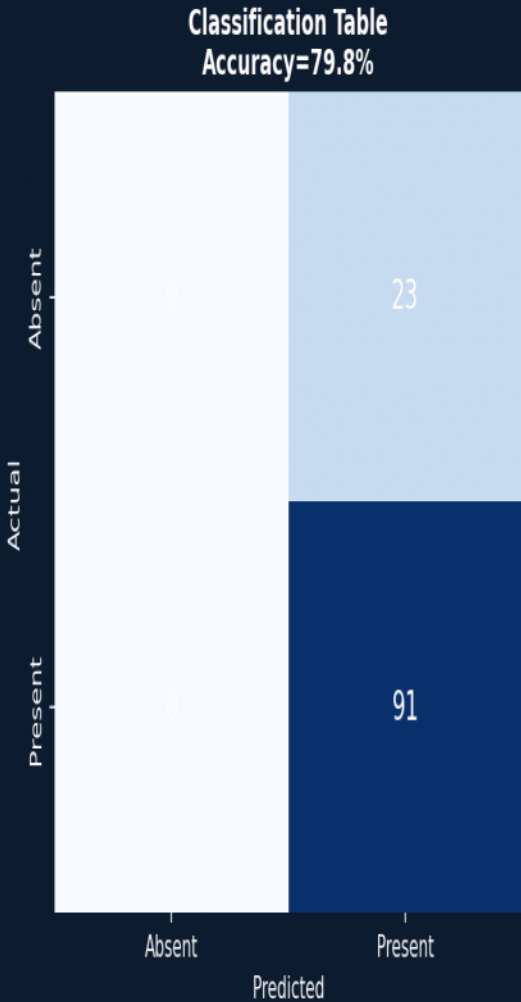
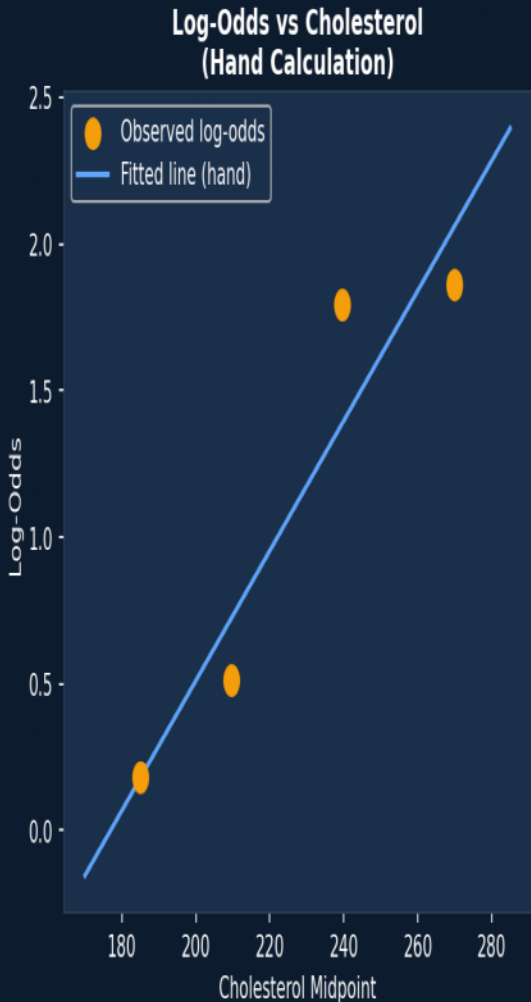
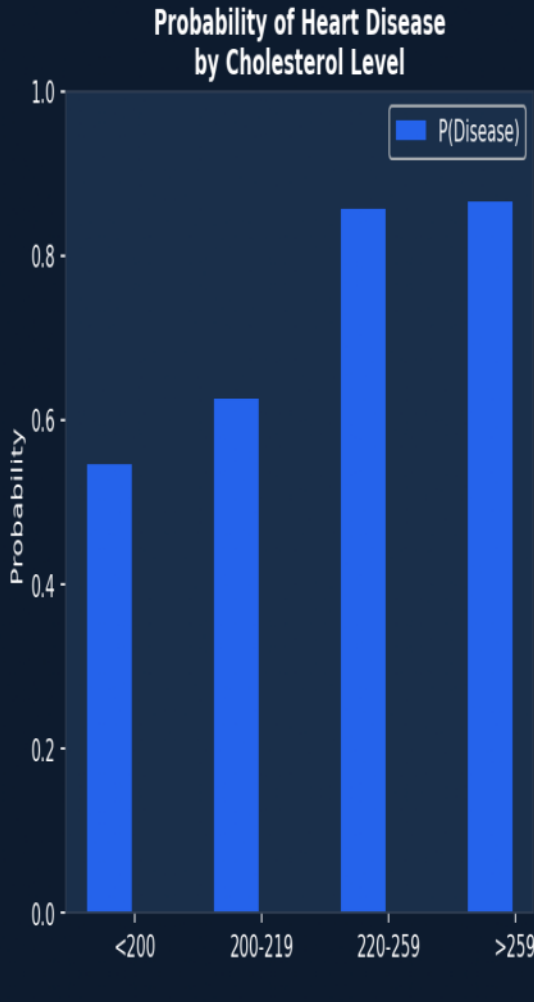
$\text{log-odds} = -3.9181 + 0.022144 \times \text{Cholesterol}$

Interpretation

- ▶ All cholesterol groups above <200 have higher disease probability.
- ▶ Log-odds increase nearly linearly with cholesterol level — clear dose-response.
- ▶ Slope > 0 confirms higher cholesterol raises heart disease risk.

Cholesterol — Software LR Results & Visualizations

(c) Software Logistic Regression with dummy variables | (d) Classification Table



Coefficient Table:

Term	Coef	p-value
Intercept	0.1823	0.7633
00-219	0.3285	0.6798
20-259	1.6094	0.0377*
259	1.6784	0.0214*

d) Classification Table Discussion

Accuracy = 79.8%

The model correctly classifies all 91 disease-present cases but misses all 23 disease-absent — high recall, zero specificity. This reflects the class imbalance (100% disease present). Classification table should be interpreted with caution: AUC is more informative.

Depression Data — LR vs Discriminant Analysis

294 subjects. Dependent variable: CASES (0=Normal, 1=Depressed where CESD>16).

84.0%

LR Accuracy

0.779

LR AUC

83.0%

LDA Accuracy

0.776

LDA AUC

Significant Predictors (LR, $p < 0.05$)

- ▶ SEX ($p=0.038$) — females significantly more likely to be depressed
- ▶ INCOME ($p=0.044$) — higher income reduces depression risk
- ▶ RELIG ($p=0.028$) — religious affiliation has protective effect
- ▶ BEDDAYS ($p=0.029$) — days spent in bed raises depression risk

LR vs LDA Comparison

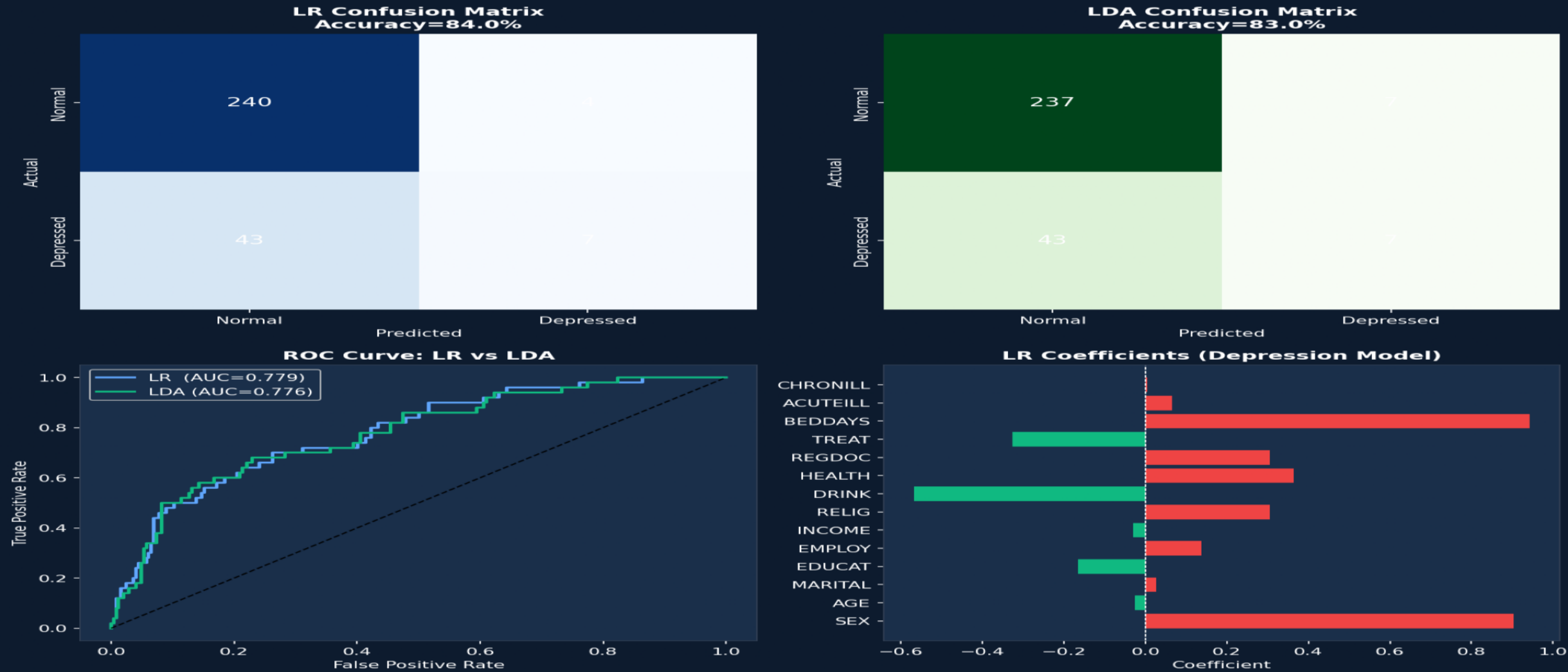
LR: Accuracy=84.0%, AUC=0.779
LDA: Accuracy=83.0%, AUC=0.776
Both methods perform similarly. LR slightly outperforms LDA on accuracy and AUC, consistent with LR's weaker distributional assumptions.

Interpretation

The logistic regression model explains depression with an overall accuracy of 84.0%. The four significant predictors align with established clinical literature: female sex, low income, lack of religious support, and health-related bed days are all established risk factors for depression. LDA matches LR closely (83.0%), suggesting the linear decision boundary is appropriate here despite slight class imbalance (244 normal, 50 depressed).

Depression — Visualizations

Confusion matrices (LR vs LDA), ROC curves, and significant coefficient bar chart.



MANOVA — Depression & Phone Data

MANOVA tests whether group means on multiple dependent variables differ simultaneously.

Part A: Depression Data (DVs = C1-C20, Group = CASES)

0.2970

Wilks' Lambda
(Depression)

32.31

Approx F(20,273)

p < .0001

MANOVA p-value

98.0%

LDA Accuracy
(C1-C20→CASES)

Interpretation — Depression MANOVA

Wilks' Lambda = 0.297 (far from 1) and F(20,273)=32.31 (p<.0001): the two groups (normal vs depressed) differ significantly on the combined set of 20 CES-D items. ALL 20 items show significant univariate F-tests (p<.0001), meaning every individual depression symptom distinguishes the groups. LDA achieves 97.96% accuracy using C1-C20.

Part B: Phone Data (DVs = A1-A6, Group = n_phones)

0.3246

Wilks' Lambda
(Phone)

30.45

Approx F(12,484)

p < .0001

MANOVA p-value

80.0%

LDA Accuracy
(A1-A6→phones)

MANOVA — Visualizations & LDA Comparison

Group mean heatmap, univariate F-statistics, attitude profiles, and MANOVA vs LDA comparison.

